THE COMPLETED NAIVELY TRIVIALIZED COTANGENT COMPLEX AS THE DERIVED LIMIT OF ITS REDUCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source gives the full Artin-Rees comparison chain and the completeness argument for (2); the compatibility of the canonical quotient maps with differentials is stated as a direct check ('a moment's thought')

Editorial note. The label lemma-NL-is-limit names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the naively trivialized cotangent complex NL of a presentation and its behaviour under base change
- the two categories of the chapter and the rings B_n = B/I^nB, A_n
- Artin-Rees for Noetherian rings
- homotopy limits Rlim of inverse systems of complexes and pro-isomorphic systems

Source reference. Original lemma 00IN: sources/stacks-chow/tags.txt, source lines 685–685. Prerequisite pointer invoked at proof line(s) 636; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00MA: sources/stacks-chow/tags.txt, source lines 816–816. Prerequisite pointer invoked at proof line(s) 662; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00S0: sources/stacks-chow/tags.txt, source lines 1022–1022. Prerequisite pointer invoked at proof line(s) 612; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08TB: sources/stacks-chow/tags.txt, source lines 11159–11159. Prerequisite pointer invoked at proof line(s) 620; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 598; it names a standard fact of the same chapter that the counted argument uses, and

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every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJQ: sources/stacks-restricted/pinned-original-source.tex, source lines 395–395. Prerequisite pointer invoked at proof line(s) 638; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 091D: sources/stacks-chow/tags.txt, source lines 11441–11441. Prerequisite pointer invoked at proof line(s) 664; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 07KZ: sources/stacks-chow/tags.txt, source lines 9674–9674. Prerequisite pointer invoked at proof line(s) 665; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AL2: sources/stacks-restricted/pinned-original-source.tex, source lines 303–303. Prerequisite pointer invoked at proof line(s) 605; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring and let $I \subset A$ be an ideal. Let B be an object of (0AL4). Then*

1. *the pro-objects $\{NL^\wedge_{B/A} \otimes_B B/I^nB\}$ and $\{NL_{B_n/A_n}\}$ of $D(B)$ are strictly isomorphic (see proof for elucidation),*
2. *$NL^\wedge_{B/A} = R\lim NL_{B_n/A_n}$ in $D(B)$.*

Here B_n and A_n are as in Section 0AL2.

Proof. The statement means the following: for every n we have a well defined complex NL_{B_n/A_n} of B_n-modules and we have transition maps $NL_{B_{n+1}/A_{n+1}} \to NL_{B_n/A_n}$. See Algebra, Section 00S0. Thus we can consider

$$\ldots \to NL_{B_3/A_3} \to NL_{B_2/A_2} \to NL_{B_1/A_1}$$

as an inverse system of complexes of B-modules and a fortiori as an inverse system in $D(B)$. Furthermore $R\lim NL_{B_n/A_n}$ is a homotopy limit of this inverse system, see Derived Categories, Section 08TB.

Choose a presentation $B = A[x_1, \ldots, x_r]^\wedge/J$. This defines presentations

$$B_n = B/I^nB = A_n[x_1, \ldots, x_r]/J_n$$

where

$$J_n = JA_n[x_1, \ldots, x_r] = J/(J \cap I^nA[x_1, \ldots, x_r]^\wedge)$$

The two term complex $J_n/J_n^2 \longrightarrow \bigoplus B_n \mathrm{d}x_i$ represents NL_{B_n/A_n}, see Algebra, Section 00S0. By Artin-Rees (Algebra, Lemma 00IN) in the Noetherian ring $A[x_1, \ldots, x_r]^\wedge$ (Lemma 0AJQ) we find a $c \geq 0$ such that we have canonical surjections

$$J/I^nJ \to J_n \to J/I^{n-c}J \to J_{n-c}, \quad n \geq c$$

for all $n \geq c$. A moment's thought shows that these maps are compatible with differentials and we obtain maps of complexes

$$NL^\wedge_{B/A} \otimes_B B/I^nB \to NL_{B_n/A_n} \to NL^\wedge_{B/A} \otimes_B B/I^{n-c}B \to NL_{B_{n-c}/A_{n-c}}$$

compatible with the transition maps of the inverse systems $\{NL^\wedge_{B/A} \otimes_B B/I^nB\}$ and $\{NL_{B_n/A_n}\}$. This proves part (1) of the lemma.

By part (1) and since pro-isomorphic systems have the same $R\lim$ in order to prove (2) it suffices to show that $NL^\wedge_{B/A}$ is equal to $R\lim NL^\wedge_{B/A} \otimes_B B/I^nB$. However, $NL^\wedge_{B/A}$ is a two term complex $M^\bullet$ of finite B-modules which are I-adically complete for example by Algebra, Lemma 00MA. Hence $M^\bullet = \lim M^\bullet/I^nM^\bullet = R\lim M^\bullet/I^nM^\bullet$, see More on Algebra, Lemma 091D and Remark 07KZ. □